

Ideation Phase

Brainstorm & Idea Prioritization

Date	24 June 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	4 Marks



Step 1: Team Gathering, Collaboration and Problem Statement Selection



Brainstorm & idea prioritization

- **Session Date:** 24 June 2026
- **Project Engine:** Smart Agricultural Production Optimization Engine (OptiCrop)
- **Target Outcome:** Evaluate and prioritize technical solutions for intelligent crop recommendation, suitability verification, and agricultural dataset analysis to secure a maximum of 4 Marks.

🕒 **10 minutes** to align on the agricultural problem statement.

🕒 **1 hour** to list, group, and map the 6 ML and system ideas.

👤 **Manikanta Kalyanam & Mannava Sri Teja**

1

Define your problem statement

🕒 **5 minutes**

PROBLEM

How might we develop an intelligent agricultural decision support system that helps farmers recommend the most suitable crop based on soil nutrients and environmental conditions, evaluate the suitability of a specific crop, and assist researchers in analyzing agricultural datasets for better farming decisions?

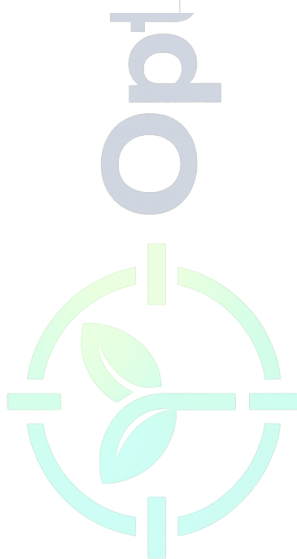
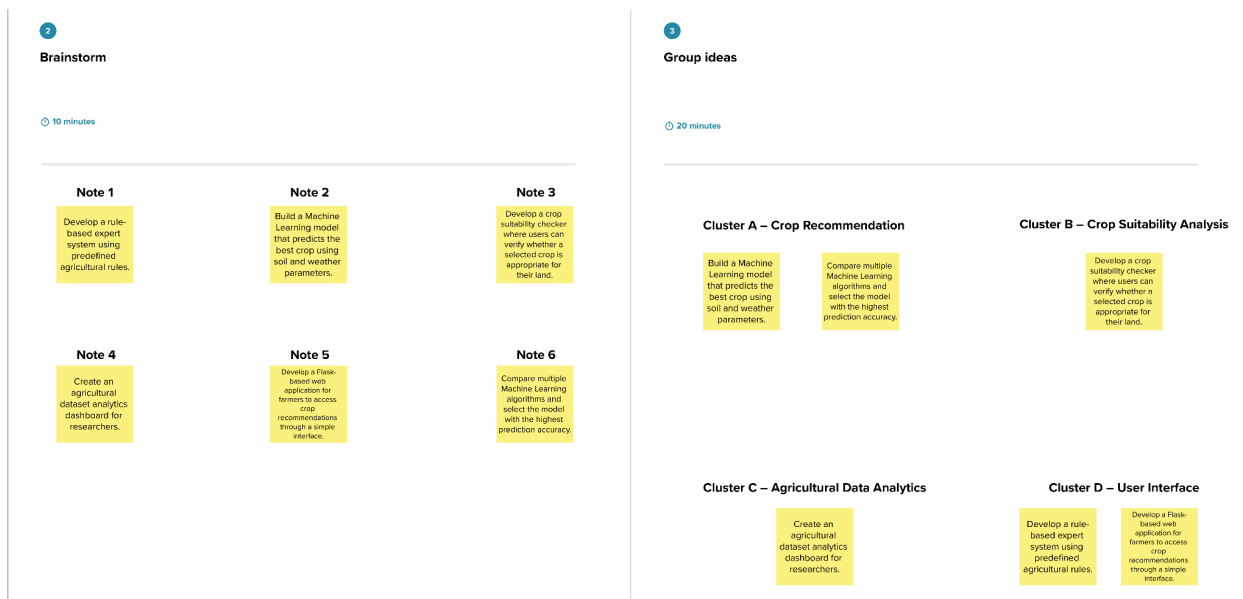


Key rules of brainstorming

To run a smooth and productive session

- | | |
|-------------------|----------------------------|
| 😊 Stay in topic. | 💡 Encourage wild ideas. |
| 👂 Defer judgment. | 👂 Listen to others. |
| 🗣️ Go for volume. | 👁️ If possible, be visual. |

Step 2: Brainstorming – Idea Listing & Grouping

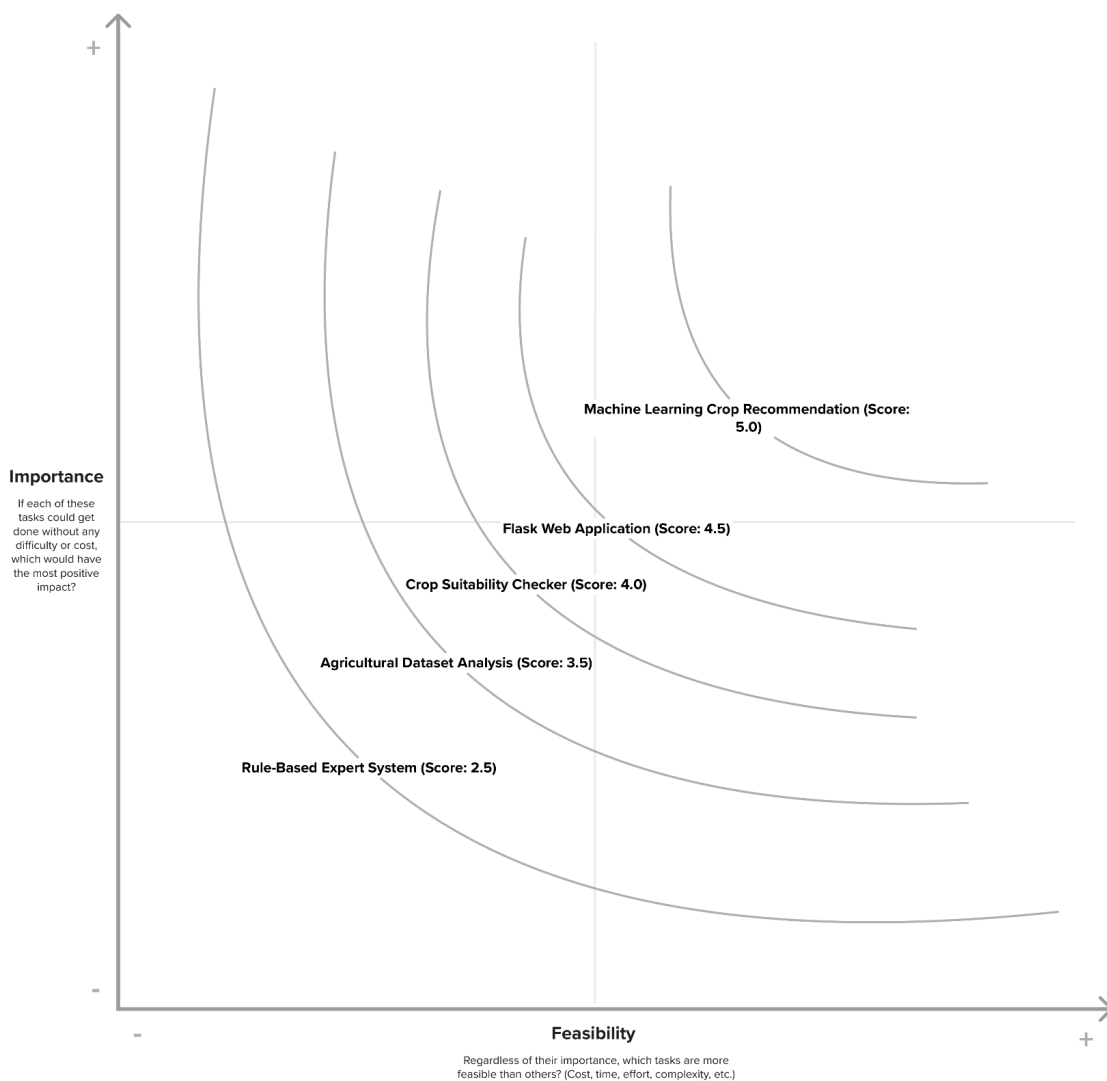


Step 3: Idea Prioritization

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Prioritize

⌚ 20 minutes



Final Prioritized Solution

The team selected the **Machine Learning-based Smart Agricultural Production Optimization Engine (OptiCrop)** integrated with a **Flask web application**. The system predicts the most suitable crop using seven agricultural parameters (Nitrogen, Phosphorus, Potassium, Temperature, Humidity, pH, and Rainfall), provides crop suitability evaluation, and supports agricultural data analysis by comparing multiple machine learning algorithms and deploying the best-performing model.

